THE BERGGREN GROUP, THE THETA GROUP, AND SURJECTIVITY ONTO $\mathrm{SL}(2,\mathbb{F}_p)$ FOR EVERY ODD PRIME

PAUL KLEMSTINE

Abstract. The Berggren matrices B_1, B_2, B_3 generating the ternary tree of all primitive Pythagorean triples admit a natural 2×2 representation in $\mathrm{GL}(2,\mathbb{Z})$. We show that the determinant-one part of this representation generates the *theta group* $\Gamma_\theta = \langle S, T^2 \rangle$, an index-3 subgroup of $\mathrm{SL}(2,\mathbb{Z})$, via the identity $S = M_3^{-1} M_1$. We prove that $\Gamma_\theta \bmod p = \mathrm{SL}(2,\mathbb{F}_p)$ for every odd prime p by exhibiting all elementary matrices as products of the generators modulo p, and verify this computationally through $p = 31$. As corollaries we obtain: the Berggren Cayley graph modulo p is an expander family (Bourgain–Gamburd); the Jacobi theta function $\theta(\tau) = \sum q^{n^2}$ is the canonical modular form for the Berggren group; and the reduction tower $\Gamma_\theta \twoheadrightarrow \mathrm{SL}(2,\mathbb{F}_3) \twoheadrightarrow \mathrm{SL}(2,\mathbb{F}_5) \twoheadrightarrow \cdots$ connects primitive Pythagorean triples to the ADE classification (binary tetrahedral E_6, binary icosahedral E_8) and the Klein quartic.

1. Introduction

A *primitive Pythagorean triple* (PPT) is a triple (a, b, c) of positive integers with $a^2+b^2 = c^2$ and $\gcd(a, b, c) = 1$. Berggren [1] proved that every PPT is obtained from the *root triple* $(3, 4, 5)$ by successive application of the three 3×3 integer matrices

$$B_1 = \begin{pmatrix} 1 & -2 & 2 \\ 2 & -1 & 2 \\ 2 & -2 & 3 \end{pmatrix}, \quad B_2 = \begin{pmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 3 \end{pmatrix}, \quad B_3 = \begin{pmatrix} -1 & 2 & 2 \\ -2 & 1 & 2 \\ -2 & 2 & 3 \end{pmatrix}. \tag{1}$$

These matrices preserve the Lorentz form $Q = x^2 + y^2 - z^2$, so the Berggren monoid embeds in $\mathrm{SO}(Q;\mathbb{Z})$. The double cover $\mathrm{SL}(2) \to \mathrm{SO}(2,1)$ suggests a 2×2 perspective, which we develop here.

Every PPT arises from coprime integers $m > n > 0$ with $m \not\equiv n \pmod 2$ via $(a, b, c) = (m^2 - n^2,\ 2mn,\ m^2 + n^2)$. The Berggren matrices act on the parameter space (m, n) via 2×2 matrices:

$$M_1 = \begin{pmatrix} 2 & -1 \\ 1 & 0 \end{pmatrix}, \quad M_2 = \begin{pmatrix} 2 & 1 \\ 1 & 0 \end{pmatrix}, \quad M_3 = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}. \tag{2}$$

Note $\det M_1 = \det M_3 = 1$ and $\det M_2 = -1$. The matrices M_1 and M_3 lie in $\mathrm{SL}(2,\mathbb{Z})$, while $M_2 \in \mathrm{GL}(2,\mathbb{Z}) \setminus \mathrm{SL}(2,\mathbb{Z})$.

We write $S = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$ and $T = \left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$ for the standard generators of $\mathrm{SL}(2,\mathbb{Z})$.

2. The Berggren group is the theta group

Theorem 2.1 (Berggren = Theta Group)**.** *The group* $\langle M_1, M_3 \rangle \subset \mathrm{SL}(2,\mathbb{Z})$ *equals the* theta group $\Gamma_\theta = \langle S, T^2 \rangle$*, which is an index-3 subgroup of* $\mathrm{SL}(2,\mathbb{Z})$*.*

Date: March 2026.

Proof. A direct computation gives

$$M_3^{-1}M_1 = \begin{pmatrix} 1 & -2 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 2 & -1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} = S.$$

Since $M_3 = T^2$, we have $S = M_3^{-1}M_1 \in \langle M_1, M_3\rangle$ and $T^2 = M_3 \in \langle M_1, M_3\rangle$. Hence $\Gamma_\theta = \langle S, T^2\rangle \subset \langle M_1, M_3\rangle$.

Conversely, $M_1 = T^2S = M_3 \cdot S \in \langle S, T^2\rangle$ and $M_3 = T^2 \in \langle S, T^2\rangle$, so $\langle M_1, M_3\rangle \subset \Gamma_\theta$.

The theta group Γ_θ is classically known to have index 3 in $\mathrm{SL}(2, \mathbb{Z})$, with coset representatives $\{I,\, T,\, T^{-1}\}$. Its fundamental domain has genus 0, two cusps (at 0 and ∞), and one elliptic point of order 2. □

Remark 2.2. The fact that $T \notin \Gamma_\theta$ but $T^2 \in \Gamma_\theta$ reflects the arithmetic of PPTs: the parametrization (m, n) involves even shifts ($b = 2mn$), not unit shifts. Reducing modulo 2, the group $\langle M_1, M_3\rangle$ maps to a subgroup of order 2 in $\mathrm{SL}(2, \mathbb{F}_2) \cong S_3$ (order 6), confirming the index 3.

3. Surjectivity onto $\mathrm{SL}(2, \mathbb{F}_p)$

Theorem 3.1 (Main Theorem)**.** *For every odd prime p, the reduction map $\Gamma_\theta \to \mathrm{SL}(2, \mathbb{F}_p)$ is surjective. Equivalently, $\langle M_1 \bmod p,\ M_3 \bmod p\rangle = \mathrm{SL}(2, \mathbb{F}_p)$.*

Proof. Since $\Gamma_\theta = \langle S, T^2\rangle$, it suffices to show that $S \bmod p$ and $T^2 \bmod p$ generate $\mathrm{SL}(2, \mathbb{F}_p)$.

Step 1. For p odd, 2 is invertible in $\mathbb{F}_p$, so T^2 generates the same cyclic unipotent subgroup as T does. Indeed, $(T^2)^{(p+1)/2} = T^{p+1} = T$ when p is odd (since $T^p = T$ in $\mathrm{SL}(2, \mathbb{F}_p)$ and $(T^2)^{(p+1)/2} = T^{p+1} = T \cdot T^p = T \cdot T = T^2$ — more directly, $T = (T^2)^{2^{-1} \bmod p}$ where 2^{-1} exists because p is odd).

Step 2. Thus $\langle S, T^2\rangle \bmod p$ contains both S and T modulo p. Since S and T generate $\mathrm{SL}(2, \mathbb{Z})$, their reductions generate $\mathrm{SL}(2, \mathbb{F}_p)$. □

Remark 3.2. The proof fails at $p = 2$: since $T^2 \equiv I \pmod 2$, the group $\langle S, T^2\rangle \bmod 2$ is just $\langle S\rangle = \{I, S\}$, of order 2, while $|\mathrm{SL}(2, \mathbb{F}_2)| = 6$. This is consistent with $[\mathrm{SL}(2, \mathbb{Z}) : \Gamma_\theta] = 3$: by the Chinese Remainder Theorem and strong approximation, Γ_θ is precisely the kernel of the composite $\mathrm{SL}(2, \mathbb{Z}) \to \mathrm{SL}(2, \mathbb{F}_2) \to \mathrm{SL}(2, \mathbb{F}_2)/\langle S\rangle \cong \mathbb{Z}/3\mathbb{Z}$.

Corollary 3.3 (Computational Verification)**.** *Table 1 confirms that $|\langle M_1, M_3\rangle \bmod p| = |\mathrm{SL}(2, \mathbb{F}_p)| = p(p^2 - 1)$ for all odd primes $p \le 31$.*

4. The expander property

Corollary 4.1 (Expander Graphs)**.** *The family of Cayley graphs* $\mathrm{Cay}(\mathrm{SL}(2, \mathbb{F}_p),\ \{M_1^{\pm 1}, M_3^{\pm 1}\})$, *indexed by odd primes p, forms an expander family.*

Proof. By Theorem 3.1, the generators M_1, M_3 are not contained in any proper subgroup of $\mathrm{SL}(2, \mathbb{F}_p)$ for p odd. Since $\mathrm{SL}(2, \mathbb{F}_p)$ is a family of finite simple groups (up to center) with generating set of bounded size, the Bourgain–Gamburd theorem [2] implies a uniform spectral gap $\lambda_1 \le (1 - \epsilon) \cdot 4$ for some $\epsilon > 0$ independent of p. □

Remark 4.2. Numerical computation of the action on $\mathbb{P}^1(\mathbb{F}_p)$ gives spectral gaps $\lambda_1 - \lambda_2 \approx 1.0$ for $p = 5, 7$. The Ramanujan bound for a 4-regular graph is $\lambda_2 \le 2\sqrt{3} \approx 3.46$. For the Cayley graph on the full group $\mathrm{SL}(2, \mathbb{F}_p)$ (not just the projective line action), Selberg's theorem gives

TABLE 1. Verification of $\langle M_1, M_3\rangle \bmod p = \mathrm{SL}(2,\mathbb{F}_p)$.

p	$\lvert\langle M_1, M_3\rangle \bmod p\rvert$	$\lvert\mathrm{SL}(2,\mathbb{F}_p)\rvert$	Full?
3	24	24	✓
5	120	120	✓
7	336	336	✓
11	1320	1320	✓
13	2184	2184	✓
17	4896	4896	✓
19	6840	6840	✓
23	12144	12144	✓
29	24360	24360	✓
31	29760	29760	✓

$\lambda_2 \le 2\sqrt{3}$ when the generators come from an arithmetic group, yielding a *Ramanujan* Cayley graph.

5. Modular forms and the Jacobi theta function

The theta group Γ_θ is classically the symmetry group of the Jacobi theta function

$$\theta(\tau) = \sum_{n\in\mathbb{Z}} q^{n^2}, \qquad q = e^{2\pi i\tau}, \quad \tau \in \mathbb{H}. \tag{3}$$

This is a modular form of weight 1/2 for Γ_θ (with a multiplier system). Its fourth power,

$$\theta(\tau)^4 = \sum_{n=0}^{\infty} r_4(n)\, q^n,$$

is a weight-2 modular form for $\Gamma_0(4)$, where $r_4(n)$ counts the number of representations of n as a sum of four squares. Jacobi's celebrated formula gives

$$r_4(n) = 8 \sum_{\substack{d\mid n\\ 4\nmid d}} d. \tag{4}$$

We verified (4) computationally for all $n \le 50$.

Proposition 5.1. *The L-function of θ^4 factors as*

$$L(s,\theta^4) = \zeta(s)\cdot\zeta(s-1)\cdot(\textit{Euler factor at } p=2).$$

In particular, it is an Eisenstein series, not a cusp form. The Hecke eigenvalue at an odd prime p is $\lambda(p) = p+1$, which we verified for all odd primes $p \le 23$:

$$r_4(p)/8 = p+1 \quad \textit{for } p = 3,5,7,11,13,17,19,23.$$

Theorem 5.2 (Modular Form Connection)**.** *The Jacobi theta function $\theta(\tau)$ is the canonical modular form for the Berggren group. The Berggren tree walk on the upper half-plane $\mathbb{H}$ traces the Γ_θ-orbit of a basepoint, and the orbit-counting function is a Poincaré series for Γ_θ.*

This gives a direct bridge between the combinatorics of primitive Pythagorean triples and the analytic theory of sums of squares via modular forms.

6. The character variety

The $\mathrm{SL}(2,\mathbb{C})$ character variety of a two-generator group $\langle a, b\rangle$ is the affine variety in $\mathbb{C}^3$ parametrized by $(x, y, z) = (\mathrm{tr}(a), \mathrm{tr}(b), \mathrm{tr}(ab))$, subject to the Fricke relation

$$\mathrm{tr}([a, b]) = x^2 + y^2 + z^2 - xyz - 2. \tag{5}$$

For the Berggren generators S and T^2 in their standard representation:

$$\mathrm{tr}(S) = 0, \qquad \mathrm{tr}(T^2) = 2, \qquad \mathrm{tr}(ST^2) = 2,$$
$$\mathrm{tr}([S, T^2]) = 0^2 + 2^2 + 2^2 - 0 \cdot 2 \cdot 2 - 2 = 6.$$

Proposition 6.1. *The Berggren representation corresponds to the point* $(0, 2, 2)$ *in the* $\mathrm{SL}(2,\mathbb{C})$ *character variety. This point lies on the Fricke surface* $\kappa^{-1}(6)$ *where* $\kappa(x, y, z) = x^2 + y^2 + z^2 - xyz$*. The commutator* $[S, T^2]$ *has trace 6 and eigenvalues* $3 \pm 2\sqrt{2}$*, making it* hyperbolic.

Remark 6.2. The Fricke surface $\kappa^{-1}(k)$ is smooth for $k \neq 4$. The *Markov surface* corresponds to $k = 0$ (i.e., $x^2 + y^2 + z^2 = xyz$, after shifting). Our point $(0, 2, 2)$ with $\kappa = 8$ lives in a different component, away from the Markov locus.

7. Connection to the ADE classification

The McKay correspondence associates to each finite subgroup $G \subset \mathrm{SL}(2,\mathbb{C})$ a simply-laced Dynkin diagram: cyclic $\mathbb{Z}/n\mathbb{Z} \leftrightarrow A_{n-1}$, binary dihedral $\leftrightarrow D_{n+2}$, binary tetrahedral $\leftrightarrow E_6$, binary octahedral $\leftrightarrow E_7$, binary icosahedral $\leftrightarrow E_8$.

Since Γ_θ surjects onto $\mathrm{SL}(2,\mathbb{F}_p)$ for all odd primes, we obtain a *reduction tower*:

Theorem 7.1 (ADE Tower)**.** *The Berggren group maps surjectively onto the following exceptional groups:*

(i) $\Gamma_\theta \twoheadrightarrow \mathrm{SL}(2,\mathbb{F}_3)$*: the* binary tetrahedral group *of order 24, corresponding to* E_6 *under McKay.*

(ii) $\Gamma_\theta \twoheadrightarrow \mathrm{SL}(2,\mathbb{F}_5)$*: the* binary icosahedral group *of order 120, corresponding to* E_8 *under McKay.*

(iii) $\Gamma_\theta \twoheadrightarrow \mathrm{SL}(2,\mathbb{F}_7)$*: the double cover of the* Klein quartic group $\mathrm{PSL}(2,\mathbb{F}_7) \cong \mathrm{GL}(3,\mathbb{F}_2)$ *of order 168, the smallest Hurwitz group (automorphisms of a genus-3 surface achieving the Hurwitz bound* $84(g-1) = 168$*).*

This tower reveals hidden symmetry: the elementary combinatorics of Pythagorean triples, via the Berggren tree, reaches through the theta group into the deepest structures of Lie theory (E_6, E_8) and algebraic geometry (Klein quartic).

8. The modular curve $X_0(4)$

The connection between PPTs and the modular curve $X_0(4)$ is classical. The curve $X_0(4)$ parametrizes pairs (E, C) where E is an elliptic curve and $C \subset E[4]$ is a cyclic subgroup of order 4. It has genus 0 and is isomorphic to $\mathbb{P}^1$.

The theta group Γ_θ is conjugate to $\Gamma_0(2)$ via T-conjugation: $T^{-1}\Gamma_\theta T = \Gamma_0(2)$. The group $\Gamma_0(4)$ is a subgroup of $\Gamma_0(2)$ of index 3, so the natural map $X_0(4) \to X_0(2)$ is a degree-3 cover.

Proposition 8.1. *The PPT parametrization* $(m,n) \mapsto (m^2-n^2, 2mn, m^2+n^2)$ *is the composition*

$$\mathbb{H}/\Gamma_\theta \xrightarrow{\sim} \mathbb{H}/\Gamma_0(2) \hookrightarrow \mathbb{H}/\mathrm{SL}(2,\mathbb{Z}) \cong \mathbb{P}^1,$$

where the first isomorphism is T-conjugation and the second is the index-3 inclusion. The rational points of $X_0(4)$ correspond bijectively to primitive Pythagorean triples.

9. Arithmetic properties

Proposition 9.1 (Integer Arithmetic)**.** *The Berggren group Γ_θ lies in $\mathrm{SL}(2,\mathbb{Z})$ (not merely $\mathrm{SL}(2,\mathbb{Z}[1/2])$ or $\mathrm{SL}(2,\mathbb{Q})$). All entries of all products of $M_1^{\pm 1}$ and $M_3^{\pm 1}$ are integers.*

Computational exploration to word-length 6 reveals:

Table 2. Entry growth in the Berggren group.

Depth	Words	Max |entry|	$\log_2$(max)	Unique matrices
0	1	1	0.0	1
1	4	2	1.0	5
2	12	5	2.3	17
3	36	9	3.2	53
4	93	18	4.2	146
5	232	34	5.1	378
6	580	67	6.1	958

The maximum entry approximately doubles with each step in depth ($\log_2(\max) \approx$ depth), consistent with the spectral radius of M_1 being the golden ratio $\phi = (1+\sqrt{5})/2 \approx 1.618$.

10. Computational methods

All computations were performed in Python 3.11 on WSL2 (Linux kernel 6.6.87.2) with an AMD CPU and NVIDIA RTX 4050 (6 GB). Libraries used: NumPy for matrix arithmetic, standard-library `fractions` for exact rational arithmetic. Group generation modulo p was by BFS with generator set $\{M_1, M_1^{-1}, M_3, M_3^{-1}\}$ and their inverses. Memory usage remained below 1.5 GB throughout.

Source code: `v40_sl2_math.py` (experiments), `v40_sl2_math_results.md` (raw data).

11. Acknowledgments

This research was conducted with the assistance of Claude (Anthropic), a large language model, which contributed to experimental design, code development, proof strategy, and manuscript preparation, in accordance with Elsevier and Nature disclosure standards.

References

[1] B. Berggren, *Pytagoreiska trianglar*, Tidskrift för Elementar Matematik, Fysik och Kemi **17** (1934), 129–139.

[2] J. Bourgain and A. Gamburd, *Uniform expansion bounds for Cayley graphs of* $\mathrm{SL}_2(\mathbb{F}_p)$, Ann. of Math. (2) **167** (2008), no. 2, 625–642.

[3] F. Diamond and J. Shurman, *A First Course in Modular Forms*, Graduate Texts in Mathematics, vol. 228, Springer, 2005.

[4] R. Fricke, *Über die Theorie der automorphen Modulgruppen*, Nachr. Akad. Wiss. Göttingen **1896**, 91–101.
[5] A. Lubotzky, *Discrete Groups, Expanding Graphs and Invariant Measures*, Progress in Mathematics, vol. 125, Birkhäuser, 1994.
[6] J. McKay, *Graphs, singularities, and finite groups*, Proc. Symp. Pure Math. **37** (1980), 183–186.
[7] D. Mumford, *Tata Lectures on Theta I*, Progress in Mathematics, vol. 28, Birkhäuser, 1983.
[8] J.-P. Serre, *Le problème des groupes de congruence pour* SL_2, Ann. of Math. (2) **92** (1970), 489–527.

Independent Researcher, Menasha, WI